

Electronic Design

The Adaptive Wearable uses a **compact microcontroller-based electronic architecture** designed to monitor environmental conditions, process sensor data, and provide immediate feedback to the user.

The main processing and control unit is an **Arduino Nano**, selected because of its compact size, sufficient digital and analog I/O, low cost, and suitability for integrating multiple sensors and output devices within a wearable prototype.

The selected electronic components are:

- **Red, yellow, and green LEDs:** Provide simple visual status indications corresponding to different environmental or system conditions. This allows the user to identify the system state quickly without continuously reading the display.
- **Buzzer:** Provides an audible warning when an important condition or threshold is detected, ensuring that alerts can still be noticed when the user is not looking at the display.
- **Push button:** Provides a simple physical input for user interaction and control of the wearable.
- **220 Ω resistors:** Used with the LEDs to limit current and protect the LEDs and microcontroller output pins.
- **I2C OLED Display:** Selected because it provides a compact and low-power visual interface while requiring only two communication lines (SDA and SCL).
- **MAX30102 Heart Rate and SpO₂ Sensor:** Selected to provide **non-invasive physiological monitoring**.
- **NEO-6M GPS Module:** Selected to provide **real-time geographical positioning**.

The components are electrically connected using **soldered connections and insulated wires**. The Arduino Nano processes the sensor inputs according to programmed threshold and decision logic and controls the OLED, LEDs, and buzzer accordingly.

The overall design prioritizes **low cost, compactness, low power requirements, component availability, and modularity**, making it suitable for a functional prototype while allowing the electronics to be upgraded in future versions.